XUANZHE (MIKE) HAN

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EDUCATION

University of California, Los Angeles (UCLA)
Master of Science in Computer Science — **Sep 2024 – Jun 2026**
Bachelor of Science in Computer Science — **Sep 2020 – Jun 2024**

SKILLS

Languages: C/C++, Python, Java, TypeScript/JavaScript, SQL
ML: PyTorch, scikit-learn
Systems/Infra: CUDA, ROS, Docker, Git, Linux
Web/Backend: React, Flask, FastAPI; Postgres, MongoDB

EXPERIENCE

Nissan North America, Inc. — Sep 2024 – Mar 2025
Systems Research Intern

- Built a configurable ROS + C++ feature-engineering pipeline for offline log replay and real-time streaming; processed **20k** log sessions (**9k** hours, **200** TB) and produced **5** routing signals (pedestrian density, bike counts, speed, obstructions, school zones).
- Contributed to an internal dashboard by implementing dataset indexing and query logic, enabling fast filtering by time range and **8** indexed fields and reducing reliance on ad hoc data-extraction scripts.
- Extended the feature/schema layer by adding new derived signals and updating data contracts as research requirements evolved.

Tap Series, LLC — Oct 2023 – Jan 2024
Web Development Intern

- Built a React + Flask analytics dashboard to parse sales and customer-interaction data and visualize trends by time, region, and demographics.
- Added interactive filtering and drill-down views, including detail pages for segment-level analysis to support internal reporting workflows.
- Integrated Twilio SendGrid to ingest **5** email event types (delivered/open/click/bounce/unsubscribe) and surface **9** campaign KPIs in the dashboard.

Huazhong University of Science and Technology — Aug 2023 – Sep 2023
Research Intern, Facial Action Unit Detection

- Adapted JAANet baselines to new facial-expression datasets by standardizing preprocessing, labels, and train/eval splits for consistent comparisons.
- Built a reproducible PyTorch model-training and scikit-learn evaluation pipeline, including data loaders, metric computation, and result aggregation scripts.
- Documented experiments (configs, dataset notes, run logs) and wrote literature review summaries to support publication preparation.

RELATED COURSEWORK

Foundations: Algorithms & Complexity, Design of Programming Languages
Systems & Networks: Operating Systems, Computer Architecture, Compilers, Parallel & Distributed Computing, Network Architecture & Protocols
AI & Data: Computer Vision, Natural Language Processing, Reinforcement Learning, On-Device LLM Deployment (llama.cpp), ML Optimization, Probabilistic Models, Data Management Systems

EXTRACURRICULAR ACTIVITIES

3D4E (3D Printing Club) 2022 – 2024
Webmaster & Team Lead *UCLA*

- Maintained the club website (3d4eatucla.github.io); shipped content/UX updates and kept documentation current to support member onboarding.
- Led biomedical and mechanical keyboard subteams; designed and delivered functional prototypes (e.g., pulse oximeter, custom macro pad), coordinating requirements, fabrication, and iteration.

PROJECTS

Ownly

- Ongoing: developing a secure decentralized collaborative workspace on the Named Data Networking (NDN) stack in Prof. Lixia Zhang's lab at UCLA.
- Integrated a Rust/OpenMLS WebAssembly module into the existing TypeScript/Go stack to manage MLS group creation, member add/join flows, commit application, and group-secret export.
- Built a 3-message MLS control plane (KeyPackage, Welcome, Commit) over the existing SVS/blob transport, deriving rotating 32-byte workspace encryption keys directly from MLS group state.
- Added recovery support by persisting 2 MLS state artifacts (storage snapshot and group ID), enabling restoration of group state across browser restarts.

DecentraDocs

- Built a browser-native real-time editor (React/Vite + Monaco + Yjs) with P2P sync over PeerJS/WebRTC; persisted state in IndexedDB for reload/offline continuity.
- Implemented a centralized control plane for identity and connectivity: Dex OIDC for canonical user IDs, WebSocket signaling for room membership/SDP+ICE exchange, and coturn STUN/TURN for NAT traversal.
- Quantified networking/reliability tradeoffs: WebRTC keep-alives every **10s** to maintain NAT bindings; single-character edits cost **1100B** (direct P2P) vs **1300B** (TURN relay); showed control-plane dependence reduces 99.9% annual uptime to **99.7%** (26.3 hrs/yr) for three required services.

Incremental Delta Snapshots for Pronghorn

- Implemented parent-linked incremental CRIU checkpoints in Pronghorn with MinIO-backed delta chains, restore-time ancestor reconstruction, and chain-safe pruning (max_chain_depth=**5**, fallback at **80%** delta/root size).
- Preserved or improved average latency on BFS/PageRank and reduced BFS max pool storage/bandwidth by **10.1%** (**1272MB→1144MB**, **5300MB→4767MB**), with minimal PageRank storage change (**0.2%**).

P-Forest

- Trained random-forest intrusion-detection models on CIC-IDS-2017 and compiled decision-tree paths into TCAM match-action rules for an RMT/CRAM simulator.
- Optimized model-to-rule compilation with deduplication and homogeneous-subtree short-circuiting, cutting RF rule count by **11–15%** and decision-tree rules by **22–27%** while preserving pipeline depth.

Dynamic Kafka

- Built a DynamicBatchProducer wrapper for Kafka that adaptively adjusts producer batch size using EMA-tracked per-message delivery latency; reconfigures batch size by flushing and respawning producers under the hood.
- Evaluated on **100,000** messages (**100B** each) and showed the controller maintains latency near a **1ms** target, dynamically shrinking batches under a **5ms** injected latency spike and increasing again once removed (with near 5–10% payload-delivery overhead from control).

On-Device LLM Deployment

- Benchmarked compact LLM inference on a Qualcomm Snapdragon device with llama.cpp, comparing LoRA fine-tuning, pruning, and 4–5 bit quantization on TruthfulQA/LongBench via ADB automation.
- Profiled throughput and power with llama-bench and Perfetto: LFM2 **1.2B** (Q4_K_M) reached **170.9 tok/s** prefill and **58.4 tok/s** decode at **2.35 W**, outperforming Llama 3.2 **1B/3B** quantized baselines on power efficiency.

CUDA RNN

- Implemented a character-level RNN in C++ with a fused CUDA forward-pass kernel for hidden-state, logits, and softmax computation.
- Improved TinyShakespeare training/inference throughput: batch size **32→128** cut per-epoch time by **~53%**, and 100-char generation dropped from **1.345s CPU** to **0.0388s CUDA**.

Cache System

- Built a FastAPI-based microservice cache evaluation harness using a mock social-media workload with **1002** synthetic user profiles (**~500B** each) and an injected **10 ms** network delay to simulate realistic service-to-DB latency.
- Evaluated eviction, prefetching, and tiered caching across **10,000-request** workloads; showed tiered caching and speculative prefetching achieved the lowest miss ratios and highest throughput on read-heavy/locality workloads, with cache-hit latency in the **microseconds** range.

Recommendation System

- Implemented PMF (ALS and SGD/Adam) and an LDA-based recommender on the Anime Recommendations Database (**73,516** users, **12,294** anime); trained on a **100,000**-rating sample with a 75/25 split and evaluated with RMSE/MAE.
- Reported test performance and training cost tradeoffs: ALS-PMF reached **~1.5–1.6** test RMSE with best **~1.43** (but overfits quickly) and took up to **27 min** on CPU; SGD-PMF trained <**10 min** on an RTX 3070Ti; LDA achieved test RMSE **1.473** / test MAE **1.137** and trained in minutes.

SlangLingo

- Built a slang-aware translation web app powered by a fine-tuned GPT-3.5 model to handle informal and idiomatic user inputs.

ResuMatch

- Built a Chrome extension that extracts LinkedIn job requirements and matches them against uploaded resumes to streamline applications.